

Identifying Influential Nodes in Epidemic Spread Using Temporal Local Variation

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Abstract—Infection spread analysis is crucial for studying and containing epidemics. In recent pandemics, a major mode of infection spread has been the movement of populations across regions. We focus on identifying regions, such as cities or states, that exert strong influence on the temporal spread of disease to neighboring regions. We utilize a graph signal-based metric, termed temporal local variation (TLV), which jointly captures spatial and temporal variations in graph signals to identify influential nodes driving epidemic spread. We apply TLV to analyze SEIR-simulated epidemic data as well as real-world COVID-19 datasets from the United States and India. Experimental results demonstrate the improved performance of TLV over global graph-based processing methods, such as graph filtering, in identifying influential nodes.

Graph signal processing, temporal local variation, spatio-temporal graph signals, SEIR model, COVID-19

I. INTRODUCTION

During epidemics such as SARS [1] and COVID-19 [2], [3], disease spread rapidly across countries, with human mobility playing a crucial role. This suggests that timely local interventions could have contained the spread. In this study, we analyze epidemic data to identify influential regions using SEIR network modeling [4], [5] and graph signal processing [6], [7], treating each region as a virtual sensor capturing time-varying infection signals.

Traditional signal processing, which assumes regular domains, has limitations when applied to signals defined over networks (i.e., irregular domains). Graph signal processing (GSP) [6], [8]–[11] has emerged as an alternative paradigm for representing and processing such signals while accounting for network structure. We analyze temporally evolving infection data with susceptible-exposed-infected-recovered (SEIR) dynamics within the GSP framework. The epidemic is modeled using reaction-diffusion dynamics [1], [4], [5], [12], [13], where agents interact within communities and move between regions. We consider a meta-population model of interconnected regions, where links represent migration and within-region interactions follow disease dynamics.

In our previous work [7], we introduced and validated the TLV metric using SIR simulations, H1N1 epidemics, traffic network data, and COVID-19 county-level analyses, along

with Monte Carlo-based parameter tuning and comparisons with existing GSP and centrality methods. In contrast, this study applies TLV within a graph signal processing framework to SEIR dynamics on small-world networks, including single and double perturbation cases, as well as real-world state-level COVID-19 data from India and the United States. TLV combines spatial and temporal information of graph signals to identify influential nodes driving spatio-temporal propagation. In epidemic settings, these nodes correspond to regions whose infection dynamics significantly influence spread to neighboring regions, such as outbreak sources or hotspots.

Prior literature on influential node identification includes graph frequency-based analysis [14], [15], spectral wavelet-based analysis for epidemic spread [16], and graph signal processing-based centrality measures [17]. While [18] focuses on reconstructing time-varying graph signals, other studies address joint temporal and spatial signal estimation on graphs [19]. In contrast, we focus on spatio-temporal signal analysis to detect influential nodes in epidemic contexts. Unlike existing approaches that primarily focus on either static graph structure or purely spatial variation, our work studies joint spatio-temporal variation of epidemic signals, enabling identification of influential regions that evolve over time. Our contributions are as follows:

- (a) Illustrate the TLV metric using a temporally evolving graph signal and analyze the effect of its weight parameter (α),
- (b) Simulate SEIR network dynamics and identify influential nodes using TLV analysis,
- (c) Process COVID-19 data from the U.S. and India to identify influential regions and validate them against reported outbreaks.

II. BACKGROUND

Let $\mathcal{G} = (\mathcal{V}, \mathbf{W})$ be an undirected graph with node set $\mathcal{V} = (1, 2, \dots, N)$ and weighted adjacency matrix $\mathbf{W}$. The graph Laplacian is defined as $\mathbf{L} = \mathbf{D} - \mathbf{W}$, where $\mathbf{D}$ is the diagonal degree matrix with entries $d_i = \sum_{j \in \mathcal{N}_i} W_{ij}$ and $\mathcal{N}_i$ are neighbors of the i th node. The graph signal at time t is $\mathbf{x}_t = [x_1, x_2, \dots, x_N]^T$ which are collected into the matrix $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T]$. The amount of variability in the graph

signal, in the context of the graph connectivity, is measured by graph signal variation. Next, we briefly review graph high pass filter and then describe the proposed metric, previously introduced in our earlier work [7].

A. Graph high pass filtering

The graph Fourier transform (GFT) [9] decomposes a graph signal $\mathbf{x}$ according to the graph topology. With the Laplacian eigen-decomposition $\mathbf{L} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T$, the GFT is $\tilde{\mathbf{x}} = \mathbf{U}^T\mathbf{x}$. Let $\mathbf{h}_{\text{HPF}}$ indicate the $M < N$ highest graph frequencies. The high-pass filtered signal [14] is $\mathbf{x}_{\text{HPF}} = \mathbf{H}_{\text{HPF}}\mathbf{x}$, where $\mathbf{H}_{\text{HPF}} = \mathbf{U}\tilde{\mathbf{H}}_{\text{HPF}}\mathbf{U}^T$ and $\tilde{\mathbf{H}}_{\text{HPF}} = \text{diag}(\tilde{\mathbf{h}}_{\text{HPF}})$. This extracts the rapidly varying components of $\mathbf{x}$ over the graph $\mathcal{G}$.

B. Total variation and local variation

The total variation (TV) [6] of a graph signal $\mathbf{x}_t$ measures its global changes over the graph,

$$\mathcal{TV}(t) = \sum_{i=1}^N \sum_{j \in \mathcal{N}_i} W_{ij} (x_t(i) - x_t(j))^2. \quad (1)$$

Although useful, due to its cumulative nature, this global measure fails to characterize the graph signal variability in local pockets of the graph. To capture local signal variability at the i th node, the local variation (LV) is defined [7] as,

$$\mathcal{LV}(i, t) = \sum_{j \in \mathcal{N}_i} W_{ij} (x_t(i) - x_t(j))^2, \quad (2)$$

$$(\mathbf{X}_{LV})_{ij} = \mathcal{LV}(i, t). \quad (3)$$

Large LV values indicate that the graph signal at that node varies significantly compared to its neighbors. Unlike high-pass filtering [14], LV captures local signal dynamics.

C. Temporal local variation

For temporally evolving graph signals, we adopt the notion of temporal local variation (TLV), which was introduced in our prior work [7], and is defined as follows:

$$\mathcal{TLV}(i, t) = \alpha \phi(i, t) (x_t(i) - x_{t-1}(i))^2 + (1 - \alpha) \sum_{j \in \mathcal{N}_i} W_{ij} (x_t(i) - x_t(j))^2, \quad (4)$$

$$\phi(i, t) = \text{sign}(x_t(i) - x_{t-1}(i)), \quad (5)$$

where $\phi(i, t)$ encodes the direction of temporal change and $\alpha \in [0, 1]$ balances temporal and neighborhood contributions and can be adjusted as desired (based on the application). TLV is capable of highlighting anomalous nodes both temporally and locally, enabling detection of influential regions in epidemic networks [7]. It is applicable to both directed and undirected graphs. For direct comparison of temporal local variation across different scenarios, we normalize it as follows.

$$\mathcal{TV}^{temp}(i, t) = \begin{cases} 0, & t = 1, \\ (x_t(i) - x_{t-1}(i))^2, & t > 1, \end{cases} \quad (6)$$

$$(\mathbf{X}_{TV})_{it} = \mathcal{TV}^{temp}(i, t). \quad (7)$$

TABLE I: Local and temporal variations for the example in Figure 1.

Nodes	Local variation			Temporal variation		
	T-1	T	T+1	T-1	T	T+1
a	0	3	3	0	1	-1
b	0	1	1	0	0	1
c	0	1	2	0	0	1
d	0	0	2	0	0	0
e	0	1	2	0	0	1

$$\mathcal{TLV}_{\text{Norm}}(i, t) = \alpha \frac{\phi(i, t) (x_t(i) - x_{t-1}(i))^2}{\max(\mathbf{X}_{TV})} + (1 - \alpha) \frac{\sum_{j \in \mathcal{N}_i} (x_t(i) - x_t(j))^2 w_{ij}}{\max(\mathbf{X}_{LV})}. \quad (8)$$

Similarly, the local variation is normalized by dividing (3) by its maximum value. These normalizations bound local variation between 0 and 1, temporal variation between -1 and 1, and subsequently temporal local variation between -1 and 2.

III. ILLUSTRATION OF LV AND TLV

In this illustration, we analyze how local variation (LV) and temporal local variation (TLV) capture fluctuations in a time-evolving graph signal. Figure 1 presents a binary-weighted graph with five nodes over three time steps, where dashed gray lines indicate temporal evolution and node color represents signal values. At time $T-1$, the signal is zero at all nodes; at T , it increases at node **a**; and at $T+1$, it decreases at node **a** while increasing at its neighbors. This scenario illustrates a spreading process, such as an epidemic, originating at a single node and propagating to adjacent nodes over time. Table I summarizes local and temporal variation components, while Table II reports normalized LV and TLV values for $\alpha \in \{0, 0.5, 0.8, 1\}$ across nodes.

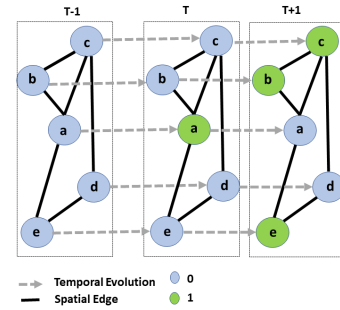


Fig. 1: Temporally evolving graph signal over three time steps.

At time T , node **a** has the highest TLV, capturing strong local and temporal variation, while node **d** remains zero due to no change in its neighborhood or temporal evolution.

At time $T+1$, the effect of α becomes evident. For $\alpha = 0$, TLV is dominated by spatial variation, making node **a** prominent due to changes in its neighbors. For $\alpha = 1$, only temporal variation is considered, reducing TLV at node **a** while increasing it at neighboring nodes. Intermediate values (0.5, 0.8) capture both effects, highlighting nodes with rising activity. Overall, TLV captures spatio-temporal dynamics via sensitivity to neighborhood structure and temporal evolution.

IV. EPIDEMIC ANALYSIS

A. Network SEIR model

We consider a network of N nodes and employ a meta-population SEIR (susceptible-exposed-infected-recovered) dynamics. Following [4], [20] we formulate the meta-population network equations. For the i^{th} node, the disease transmission rate and the average recovery rate are denoted by β_i and γ_i , respectively, while σ_i represents the inverse of the average latent period. Population migration is modeled through the diffusive term $\sum_{j \in \mathcal{N}_i} W_{ij}(X_j - X_i)$. The strength of the migration is determined by κ . To solve the ODE equations, we utilised Runge-Kutta (RK4) [21] method with adaptive step size.

Nodes	TLV $\alpha = 0$		TLV $\alpha = 0.5$		TLV $\alpha = 0.8$		TLV $\alpha = 1$	
	Local=LV		T	T+1	T	T+1	T	T+1
	T	T+1						
a	1	1	1	0	1	-0.6	1	-1
b	0.33	0.33	0.167	0.667	0.067	0.867	0	1
c	0.33	0.67	0.167	0.833	0.067	0.93	0	1
d	0	0.67	0	0.33	0	0.13	0	0
e	0.33	0.67	0.167	0.833	0.067	0.93	0	1

TABLE II: Normalized LV and TLV for the example in Fig.1.

B. Methodology for identifying influential nodes

Given a graph $\mathcal{G}$ with an associated graph signal $\mathbf{X}$, our goal is to identify the nodes that critically influence the temporal evolution of the signal. Following the framework developed in our previous work [7], we identify nodes that significantly influence the temporal evolution of a graph signal $\mathbf{X}$. As outlined in Algorithm 1 the method takes the graph

Algorithm 1 Online identification of influential nodes

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1: for  $t = r, r + 1, 2, \dots$  do
2:    $\bar{\mathbf{X}} = \mathbf{X}(:, t - r + 1 : t)$ 
3:   Compute  $\bar{\mathbf{X}}_{var}$  using HPF or (3) or (8)
4:    $\bar{\mathbf{x}}_{sort} = \text{sort}(\bar{\mathbf{X}}_{var}(:, r), \text{'descend'})$ 
5:   top  $p\%$  nodes of  $\bar{\mathbf{x}}_{sort}$  are influential nodes at time
   step  $t$ 
6: end for

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$\mathcal{G} = (\mathcal{V}, \mathbf{W})$ and signal $\mathbf{X}$ as inputs. At each time t , a temporal window of length r (from $t - r + 1$ to t) is extracted as $\bar{\mathbf{X}}$. The signal variation $\bar{\mathbf{X}}_{var}$ is computed using one of HPF $_{\mathcal{G}}$, LV, or TLV, and the variation at the most recent time $\bar{\mathbf{X}}_{var}(:, r)$ is sorted. The top $p\%$ nodes are then selected as influential at time t . For comparison, a baseline “Max processing” method ranks nodes by their raw infection values at time t , selecting the top $p\%$. All the methods identify the influential nodes in ‘real-time’, using recent historical data, allowing for timely local intervention.

V. SIMULATIONS AND RESULTS

We evaluate the proposed framework against representative graph signal processing methods and classical graph centrality-based approaches. Comprehensive comparative experiments across scenarios and datasets, including ranking-based evaluations, are provided in [7].

A. Graph and epidemic data generation

Graph construction: We generated a weighted small-world network following the Watts–Strogatz model [22], with $N = 50$ nodes uniformly placed on a unit circle and each node connected to its $k = 4$ nearest neighbors as shown in Figure 2a. Edges were rewired with probability $\gamma = 0.1$. Edge weights were assigned using a Gaussian kernel based on Euclidean distance between nodes: $W_{ij} = \exp(-\|\mathbf{u}_i - \mathbf{u}_j\|^2 / \tau^2)$, with $\tau = 0.5$, where $\mathbf{u}_i$ and $\mathbf{u}_j$ denote spatial coordinates of nodes.

Additional experiments involving multiple graph topologies, source nodes, and repeated Monte Carlo trials are provided in our prior work [7], demonstrating the generality and robustness of the proposed framework.

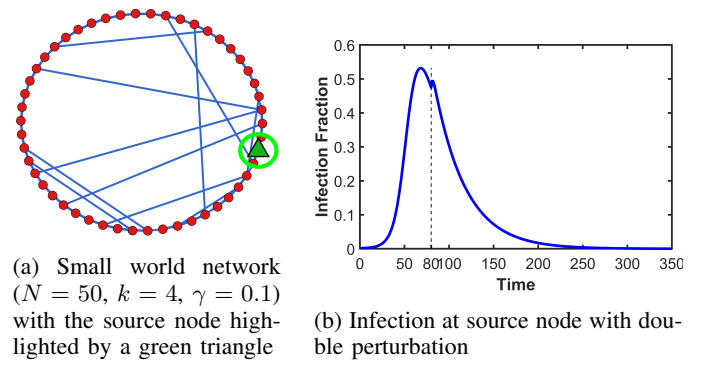


Fig. 2: Illustration of infection spread: (a) network topology with source node and (b) temporal infection at the source under double perturbations

Data generation: We simulate the epidemic starting from a randomly selected graph node (triangle marker in Figure 2a). We introduce a secondary perturbation at the initial infected node with $\beta = 0.7$. Once the node reaches its infection peak as shown in Figure 2b, σ is temporarily increased from 0.1 to 0.4 for a finite duration. This occurs at $t = 80$ for high coupling $\kappa = 0.1$. Figure 2b shows the temporal infection dynamics of the source node over 350 time steps.

Data processing and plotting: Given the graph $\mathcal{G}$ and graph signal $\mathbf{X}$, we can carry out epidemic analysis using Algorithm 1 with results presented in Figure 3 and 5. Various stages of the epidemic are presented with triangle indicating the infected source node. The identified influential nodes are highlighted in red for visualization. Data is plotted on a logarithmic scale.

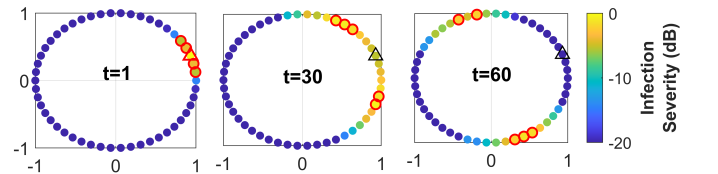


Fig. 3: TLV-based visualization shows disease spread via the color scale, with influential nodes highlighted in red.

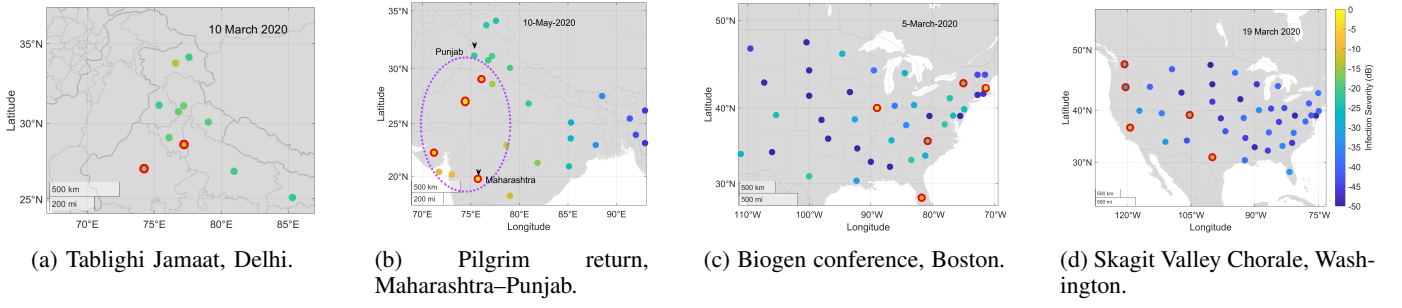


Fig. 4: Influential regions identified by Algorithm 1 using TLV metric: (a) Tablighi Jamaat in Delhi, (b) Pilgrim return from Maharashtra to Punjab, (c) Biogen conference in Boston, and (d) Skagit Valley Chorale in Washington.

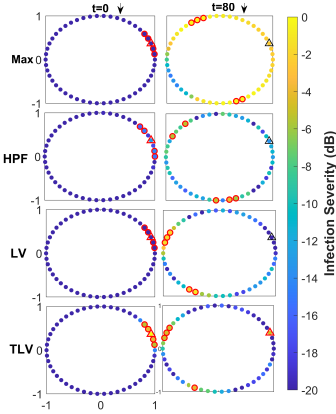


Fig. 5: Identification of single and double perturbation at infected source node (in triangle) for $\kappa = 0.1$.

B. Perturbation analysis

Our objective here is to understand how the disease spreads from a single infected node to the entire network. Results of Algorithm 1 using TLV are shown in Fig. 3 at multiple time steps. TLV identifies the infected source node at $t = 1$ and shows epidemic spread across the network. The node color indicates TLV value, revealing infection severity across nodes, while red nodes indicate influential nodes. Figure 5 shows the results of Max processing, HPF (top 25% of frequencies), LV, and TLV. All methods identify the infection source node (in triangle) at time $t = 0$, but only TLV identifies the presence of a second perturbation at this node at $t = 80$.

Nodes with spatially varying signals are detected by LV, while TLV identifies nodes with spatio-temporal variations. Many influential nodes overlap between LV and TLV, but TLV also captures temporal evolution, leading to some differences. Although Max processing highlights nodes with the highest infection, it does not fully capture infection dynamics.

VI. COVID-19 DATA ANALYSIS

We demonstrate the effectiveness of TLV-based processing on real-world COVID-19 datasets from India and the U.S., identifying influential regions using population-normalized confirmed cases as graph signals and mobility-based state graphs. The analysis relies on reported data without explicit correction for false positives or seroconversion effects.

For India, publicly available COVID-19 confirmed cases [23] across $N = 36$ states and union territories were normalized by respective state populations [24]. An inter-state road network $\mathcal{G}$ was constructed using distance-based weights [7], assigning higher weights to closer states with threshold $\vartheta = 1000$ km and $\tau = 500$. Applying Algorithm 1 using TLV ($\alpha = 0.6$, $r = 10$, $p = 10\%$), influential states were identified. Figure 4a shows Delhi detected as influential, corresponding to the Tablighi Jamaat congregation [25] from March 1–20, 2020. Figure 4b shows the interstate pilgrim [26] return from Maharashtra to Punjab on May 2, 2020, where TLV identifies influential nodes along the route, with a violet ellipse highlighting the infection pathway.

For the U.S., publicly available COVID-19 confirmed cases [27] across $N = 55$ states were normalized by population [28]. A directed graph $\mathcal{G}$ was built using airline data from the Bureau of Transportation Statistics [29], with weights given by passenger numbers between states. Applying TLV ($\alpha = 0.5$, $r = 10$, $p = 10\%$) identifies influential states and multiple super-spreader events. Figure 4c shows the Biogen conference [30] in Boston identified on March 5, 2020, while Figure 4d shows the Skagit Valley Chorale event [31] in Washington, detected on March 19, 2020, reflecting the incubation period. While TLV identified several other events, only a subset is presented here due to space constraints. The results indicate temporal consistency between high TLV regions and reported outbreak events, though they should be interpreted as associations rather than causal identification.

VII. CONCLUSION AND FUTURE WORK

We examined disease spreading patterns from infected source nodes using graph signal variation. A low-complexity spatio-temporal variation metric from our prior work [7], TLV, was used to identify influential nodes. Visualization of local and temporal variation highlights hot-spots that enable rapid disease spread, providing actionable insights for policymakers to classify regions by infection severity and implement preventive measures. Additional experiments show that TLV accurately identifies influential nodes and outperforms existing approaches. Future work includes developing a theoretical framework for these metrics and extending the approach to more complex spatio-temporal data and infections.

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